Digital Deception Detection Suite

Project Proposal for UCS503P (Software Engineering)
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Repository: `https://github.com/harmilan141/Digital-Deception-Detection-Suite`

1 Higher-order goal *. . . secondary framing*

This project aims to make everyday digital interactions safer by reducing the time and expertise a user needs to recognize deceptive content, including phishing messages, fake/malicious URLs, fabricated news articles, and manipulated (deepfake) media. The immediate engineering objective is to develop a measurable and deployable software solution for *multi-modal deception detection and risk scoring across text, links, and media shared with a user.*

2 Time-to-value *. . . primary heuristic*

- We will accelerate value delivery by rapidly prototyping a single detection pipeline (e.g., phishing/URL scoring) end-to-end and validating it with real inputs within each short development cycle.
- We will favor continuous feedback over premature optimization, using weekly iterations and early CI-backed automation to minimize deployment overhead.
- Initial success will be measured through tangible, testable outcomes from the first iteration (a working detector with a reported accuracy), with subsequent releases driven by validated user feedback.

3 Problem Statement

Everyday users are increasingly exposed to deceptive digital content that is difficult to identify without technical expertise: phishing emails/SMS impersonating banks or services, shortened or spoofed URLs leading to malicious sites, fabricated or misleading news articles, and AI-generated deepfake images, audio, and video used for scams or misinformation. Common operational pain points include:

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- **No unified check:** Users must rely on separate, unfamiliar tools to verify suspicious links, messages, and media files independently.
- **Low detection literacy:** Most non-technical users cannot reliably distinguish sophisticated phishing attempts or synthetic media from genuine content.
- **Delayed or absent warnings:** Deceptive content is often identified only after harm occurs (e.g., credential theft or unauthorized financial transactions).
- **Fragmented signals:** Disparate deception indicators (domain age, sender reputation, linguistic markers, manipulation artifacts) are not aggregated into an actionable, unified score.

Why this matters now (contextual relevance): Generative AI tools have lowered the technical and financial barrier to creating hyper-realistic deceptive text, cloned voices, and forged videos. A lightweight, explainable detection layer that non-expert users can query before interacting with untrusted digital artifacts directly mitigates modern social-engineering attacks.

4 Proposed Solution *. . . Software Proposition*

4.1 Overview

A unified web platform providing multi-modal analysis:

- **Text/message analyzer:** Evaluates text payloads for social-engineering indicators (urgency, credential/financial requests, known malicious patterns).
- **URL/link risk checker:** Evaluates links using domain-reputation feeds, lexical characteristics (length, character entropy, typosquatting), and redirect traces.
- **Media authenticity checker:** Analyzes images, audio, and short video clips using pretrained computer-vision and audio manipulation detection models.
- **Unified risk score:** Aggregates sub-signals into an interpretable risk level (*Low*, *Medium*, *High*) with clear textual rationale.
- **Report/history log:** Retains user-submitted scan logs for reference and offline benchmark evaluation.

4.2 Core workflow

1. **Submission:** User submits raw text, a target URL, or a media file.
2. **Routing:** The gateway routes the payload to the relevant analyzer microservice(s).
3. **Feature Extraction:** Analyzers extract lexical, structural, or neural features and output sub-scores with human-readable evidence.
4. **Aggregation:** The unified scoring engine applies transparent heuristic weights to generate an overall risk score and summary.
5. **Display:** The interface presents the verdict, key indicators, and clear remediation guidance.
6. **Logging:** The transaction metadata is persisted for tracking and testing.

4.3 Operational constraints for everyday users

- Maintain low latency by executing compute-heavy inference server-side, keeping client requirements minimal.
- Provide explainable evidence over black-box outputs to reinforce user awareness.
- Integrate pre-trained open-source weights and third-party threat feeds to ensure implementation viability within academic deadlines.
- Fail safely (e.g., returning *"Unable to verify"* rather than an inaccurate *"Safe"* status) during third-party service degradation.

5 Solution Approach *. . . Engineering focus*

This project prioritizes dependable system integration, clean API orchestration, and low-latency serving over novel model architecture research.

5.1 Detection logic *. . . non-research*

- **Text Analyzer:** Regex-based heuristic filters combined with a pre-trained transformer model fine-tuned for spam/phishing classification.
- **URL Analyzer:** Feature extraction (homoglyphs, entropy, path depth) paired with domain-age and reputation querying via public threat APIs.
- **Media Analyzer:** Open-source pre-trained deepfake detection networks (e.g., spatial-frequency artifact detectors, audio spectral analysis).
- **Scoring Integration:** Weighted rule-based aggregation function ensuring explainability without second-tier black-box latency.

5.2 Web architecture *. . . web-based solution*

- **Frontend:** Responsive single-page application (React / Tailwind CSS).
- **Backend API:** Stateless REST backend (FastAPI / Node.js) handling validation, job dispatch, scoring, and authentication.
- **Database:** Relational database (PostgreSQL / SQLite) for scan metadata, user management, and evaluation records.

5.3 CI/CD and fast delivery enabler

- Automated unit and integration testing running against synthetic phishing/clean datasets on every pull request.
- Containerized deployment pipelines (Docker, GitHub Actions) pushing automated builds to staging.

6 Evaluation Criteria *. . . measurable and attributable*

6.1 Primary evaluation metric

Detection Accuracy per Modality: Percentage of correctly classified items on held-out benchmark datasets.

- **Target:** $\geq 85\%$ accuracy across text, URL, and media test sets.
- **Attribution:** Verified against public datasets (e.g., Nazario Phishing Corpus, Kaggle Phishing URL dataset) and a curated internal test suite.

6.2 Secondary evaluation metrics

- **False Positive Rate (FPR):** Maintained below 5% on benign data to prevent alert fatigue.
- **Explanation Coverage:** Target 100% evidence reporting on all items flagged as Medium or High risk.
- **Response Latency:** $\leq 1.5s$ for text/URL checks; $\leq 5.0s$ for media analysis.
- **User Trust/Actionability:** Measured via post-pilot survey scores on verdict clarity.

6.3 Pilot validation plan *. . . high-level*

- Pilot evaluation involving 10–15 student users testing known-safe and known-deceptive samples.
- Quantitative evaluation of system throughput, accuracy, and failure rate under active testing.

7 Scalability *. . . with foresight*

1. **System Scalability:** Decoupled frontend/backend architectures with asynchronous background workers (e.g., Celery/Redis) handling compute-intensive media inference.
2. **Operational Deployability:** Fully containerized runtime (Docker) deployable to cloud-hosted environments with centralized environment management.

8 Engine Availability Heuristic

- **Web Framework:** FastAPI (Python) or Express.js (Node.js) with React frontend.
- **Pre-trained Models:** Hugging Face Transformers for text; open-source PyTorch checkpoints for deepfake image/audio inference.
- **Threat Intelligence:** Google Safe Browsing API, VirusTotal API, and Whois lookup packages.
- **Database & CI/CD:** PostgreSQL database, PyTest/Jest testing frameworks, and GitHub Actions CI/CD workflows.

9 Project Scope and Deliverables *. . . iteration-friendly*

9.1 Initial deliverable *. . . first iteration*

- Core text and URL analysis modules with rule-based heuristics and baseline classification.
- Unified scoring mechanism with explanatory output for text/link checks.
- Automated CI pipeline with unit test coverage and staging deployment.

9.2 Subsequent deliverables

- Media manipulation checking module (image/audio deepfake detection).
- User history dashboard and exportable scan reports.
- Refined score weighting tuned on pilot validation findings.

10 Risks and Mitigations

- **False Sense of Security:** Present probabilistic risk levels with clear disclaimers rather than binary safe/unsafe guarantees.
- **External API Outages:** Implement timeout fallbacks returning graceful *"Service Unavailable"* flags without crashing the pipeline.
- **User Privacy:** Enforce data retention limits, strip raw personal identifiers, and provide user-initiated log purging.

11 Summary

This proposal presents an explainable, multi-modal deception detection platform designed for practical deployment and end-user accessibility. By utilizing established open-source models, iterative CI/CD practices, and transparent scoring metrics, the project delivers rapid time-to-value while fulfilling all software engineering milestones for UCS503P.